

Module 2 Assignment: Formulating a Design Plan

585.617 Rehabilitation Engineering | Module 2 Assignment | [Your Name] | [Date]

Design Prompt: Retractable Sun and Rain Awning for a Powered Wheelchair

Part A — Need Statement

Need statement: *A way for full-time powered wheelchair users with limited hand function to control their exposure to sun and precipitation, independently and on demand, so that weather does not determine whether they leave home.*

Supporting rationale. Two characteristics of this population make weather exposure a functional problem rather than a comfort problem. First, individuals with spinal cord injury at or above roughly T6 lose sympathetic control of sudomotor and vasomotor function below the lesion, so they cannot dissipate heat by sweating over most of the body surface and are at elevated risk of hyperthermia under sustained sun exposure [4]. Second, the joystick, controller, and connector housings are the most moisture-vulnerable subsystem of a powered chair, so users routinely forgo travel when rain is forecast. In ICF terms the result is a participation restriction produced by an environmental factor rather than by the underlying impairment — precisely the class of problem rehabilitation engineering exists to address. Users also cannot reposition into shelter as quickly as ambulatory people can, and many have limited hand function, so any solution requiring a caregiver to erect it has already failed the need.

Note on solution neutrality. The task as assigned (“add a retractable awning”) names a solution, not a need. Following Dieter and Schmid [3], the statement above is written solution-neutrally so that concept generation is not foreclosed before the design input phase: a canopy is one embodiment, but a deployable hood, an integrated garment, or a sealed-controller upgrade paired with weather alerting may screen better against the same need. The plan below carries the retractable awning as the presumptive concept while preserving a genuine concept-screening gate in Phase 2.

Part B — Design Plan (FDA Waterfall)

This plan follows the design control waterfall of the FDA *Design Control Guidance for Medical Device Manufacturers* [1], as required by 21 CFR 820.30 [2]. Because the awning is an accessory to an already-marketed Class II powered wheelchair, it is executed as a design change carried through the full waterfall, with a change assessment against the existing 510(k) as a Phase 1 deliverable, and all outputs filed in the Design History File.

Critical User Needs and Traced Design Inputs

Two critical user needs govern the project. **UN-1:** the user must be able to deploy and stow the protection without assistance, from the seated position, with impaired hand function and without reaching overhead. **UN-2:** the protection must actually keep sun and water off both the occupant and the chair’s electronics, without changing where the user can go or how stable the chair feels. Each is traced to measurable design inputs:

- **DI-1.1** (from UN-1) — A full deploy or stow cycle shall complete in ≤ 10 s, initiated by a single control requiring ≤ 22 N (5 lbf) of force and no tight grasping, pinching, or twisting of the wrist, per ADA §309.4 [6].
- **DI-1.2** (from UN-1) — The control shall sit within the seated forward and side reach envelope, require no reach above shoulder height, and be operable single-handed by a user with C6 tetraplegia (tenodesis grasp, no active finger flexion).
- **DI-2.1** (from UN-2) — Stowed, the accessory shall not alter the chair’s ISO 7176-5 overall length, width, or turning envelope. Deployed, occupied height shall not exceed 1500 mm, width shall not exceed the base chair’s width, and the chair shall meet its baseline ISO 7176-1 and ISO 7176-2 stability limits [5].
- **DI-2.2** (from UN-2) — Canopy fabric shall provide $UPF \geq 50$ and hydrostatic head ≥ 1000 mm (ISO 811). Deployed coverage shall shade the occupant’s head, shoulders, and joystick at solar elevations of 30° to 90°, with no loss of controller function under IPX4 spray.

Design review structure. Every gate is chaired by the project engineer and attended by product management, a clinical advisor (PT/OT or ATP), regulatory affairs, quality, manufacturing, and at least one reviewer with no direct responsibility for the stage under review,

as 21 CFR 820.30(e) requires. Materials circulate five business days in advance, action items are recorded with owners and due dates, no review closes with an open safety-related action, and minutes and the exit decision are filed in the DHF. What changes from gate to gate is the question asked, the material presented, and the exit criteria.

Phase 0 — User Needs (Weeks 1-8)

Tasks: interview 10–15 powered chair users spanning diagnoses and hand function plus three to five clinicians; observe use in situ; draft the need statement and a weighted needs list; build a preliminary hazard list. *Information to gather:* how often and at what cost users forgo travel in weather; caregiver availability; existing workarounds; seated anthropometric and reach data; competitive and patent landscape; warranty history for moisture-related controller failures. *Review DR-1* answers “should we build this?” and presents the need statement, user research summary, patent landscape, preliminary hazard analysis, and business case. *Exit criteria:* needs list approved and signed; every need traceable to primary research rather than assumption; no unresolved showstopper hazard; funded go decision.

Phase 1 — Design Input (Weeks 9-18)

Tasks: translate every user need into at least one measurable design input; draft the Product Design Specification; open the risk management file [8]; complete the 510(k) change assessment; write the verification strategy and the human factors plan [9]. *Information to gather:* the applicable ISO 7176 parts and their acceptance limits [5]; IEC Guide 71 provisions on reach, force, and legibility for older and disabled users [7]; the base chair’s battery and range budget and controller bus capacity. *Review DR-2* answers “will it work?” and presents the PDS, the traceability matrix, the risk analysis, the verification strategy, and the regulatory pathway determination. *Exit criteria:* 100% of user needs traced to at least one design input; every input stated in measurable terms with a named verification method; every hazard assigned a risk control.

Phase 2 — Design Process (Weeks 19-34)

Tasks: generate at least six concepts; screen them with a Pugh matrix against the weighted needs; prototype the top two; run the analyses below; freeze the design. *Information to gather:* supplier actuator data, fabric test certificates, and the measured center of gravity of the loaded base chair. Two technical aspects dominate this phase:

- **TA-1 — Wind loading and center-of-gravity shift.** The canopy is a sail mounted above the occupant’s CG, so it adds both a lateral force and a long moment arm. Sizing to $F = \frac{1}{2}\rho v^2 C_d A$ with $A \approx 0.5 \text{ m}^2$, $C_d \approx 1.2$ and $\rho = 1.225 \text{ kg/m}^3$ gives roughly 37 N at 10 m/s (22 mph) applied near 1.4 m height — a tipping moment large enough to threaten the ISO 7176-1 stability margin. This analysis sets the auto-stow threshold and drives mast placement, so it belongs on the critical path rather than in a late design check.
- **TA-2 — Material selection under competing objectives.** Canopy ribs are traded among 6061-T6 aluminium, glass-filled PA66, and pultruded GFRP against stiffness-to-weight, UV resistance, cost, and recyclability; fabric is traded between solution-dyed acrylic and PU-coated polyester against UPF retention, hydrostatic head, and packed volume. Green engineering enters as a stated screening criterion rather than an afterthought: the screen favours a mono-material, mechanically disassemblable canopy and excludes PVC coatings and PFAS-based water repellents, accepting a weight penalty for end-of-life recyclability.

Anthropometrics run through both: seated eye height sets canopy height, ADA §307.4’s 80 in (2030 mm) protruding-object limit and the typical 32 in doorway clear width bound the deployed envelope [6], and seated reach envelopes fix control placement per DI-1.2. *Review DR-2b* presents the Pugh matrix and prototype data. *Exit criteria:* one concept selected with documented rationale; feasibility demonstrated against DI-1.1, DI-2.1, and DI-2.2; no open high-severity risk lacking a control.

Phase 3 — Design Output (Weeks 31-42, overlapping)

Tasks: release drawings, specifications, BOM, actuator firmware, labelling and instructions for use; complete the DFMEA; draft the Device Master Record; write and approve every verification and validation protocol; submit the home-use study protocol for IRB review. *Information to gather:* supplier qualification data and tooling capability studies. *Review DR-3* answers “will it be successful?” and presents the complete output package, the DFMEA, essential-output

identification, and the approved protocols. *Exit criteria*: every design output traces to a design input; essential outputs identified for manufacturing acceptance; all protocols approved **before** any testing begins.

Phase 4 — Verification (Weeks 43-54)

Verification asks whether the device was designed right — whether outputs meet inputs. Two plans carry the critical inputs:

- **VER-1** (verifies DI-1.1 and DI-1.2) — Instrumented actuation testing on 25 production-equivalent units: a force gauge measures peak control force and a timer measures full cycles across the specified temperature range, with a representative-user panel confirming single-handed operation. *Acceptance*: peak force ≤ 22 N and cycle time ≤ 10 s on 100% of units. A 10,000-cycle durability run built on the ISO 7176-8 fatigue philosophy [5] follows, with force and time re-measured at completion and no loss of function permitted.
- **VER-2** (verifies DI-2.1 and DI-2.2) — Dimensional verification per ISO 7176-5 stowed and deployed, then static and dynamic stability per ISO 7176-1 and ISO 7176-2 with the awning deployed across the rated occupant mass range. Lateral load is then applied by calibrated fan in 2 m/s increments to confirm the auto-stow threshold actuates below the measured tipping condition. Fabric UPF and hydrostatic head (ISO 811) are tested on three production lots, and an IPX4 spray test runs with the controller powered. *Acceptance*: stowed envelope unchanged; baseline stability limits met; auto-stow triggers on 20 of 20 trials; no controller fault.

Phase 5 — Validation (Weeks 55-68)

Validation asks whether the right device was designed — whether user needs are met in actual use. The plan is a human factors validation with 15 representative users [9] followed by a four-week in-home use study on production-equivalent units, instrumented with the survey below and QUEST 2.0 [10]:

- **VAL-Q1** (validates UN-1) — “Over the past week, on how many occasions did you open or close the awning, and on how many of those did you need another person’s help?” followed by the five-point agreement item, “I was able to open and close the awning by myself, quickly enough to matter when the weather changed.” *Acceptance*: at least 90% of deployments unassisted and at least 80% of participants agreeing or strongly agreeing.
- **VAL-Q2** (validates UN-2) — “Did the awning stop you from going anywhere you would normally go — through a doorway, into an elevator, onto a ramp, or into your vehicle? If yes, describe what happened.” paired with, “Did you at any time feel unsteady, or that the chair might tip, while the awning was open?” *Acceptance*: no access exclusions attributable to the accessory, and no reports of perceived instability that analysis traces to a real loss of stability margin.

Reviews DR-4 and DR-5 present verification reports and deviations, the closed traceability matrix, the human factors validation report, use-study results, and the residual risk assessment. *Exit criteria*: all inputs verified; all needs validated in actual or simulated use; residual risk acceptable against benefit.

Phase 6 — Design Transfer and Release (Weeks 63-72, overlapping)

Process validation on canopy sewing and rib forming, operator training, first-article inspection, final DHF review and design release signature, then commercial launch at Week 72.

Schedule, Milestones, and Critical Path

The seven phases run to roughly 72 weeks, or 16 to 17 months. Milestones close each phase: M1 user needs baselined at Week 8, M2 requirements and PDS baselined at Week 18, M3 concept selected at Week 26 and design frozen at Week 34, M4 outputs released at Week 42, M5 verification complete at Week 54, M6 validation complete at Week 68, and M7 design release at Week 72. The **critical path** runs from user needs through the wind-load and stability analysis of TA-1 — which sets the auto-stow requirement and therefore the actuator specification — into actuator procurement at a 12 to 14 week lead, then output release, then the 10,000-cycle durability run, roughly six weeks of continuous testing that cannot be compressed, and finally validation. Two long-lead items start early: injection tooling for the rib and hub components at 12 to 16 weeks, and IRB approval for the home-use study, submitted during Phase 3 because an

eight to ten week review would otherwise land on the critical path. Four weeks of float in Phase 2 absorbs a failed concept screen, the likeliest source of schedule loss.

References

- [1] U.S. Food and Drug Administration, *Design Control Guidance for Medical Device Manufacturers*. Rockville, MD: Center for Devices and Radiological Health, March 11, 1997.
- [2] *Code of Federal Regulations*, Title 21, Part 820.30 — Design Controls. U.S. Government Publishing Office.
- [3] G. E. Dieter and L. C. Schmid, *Engineering Design*, 5th ed. New York: McGraw-Hill, 2013.
- [4] R. A. Cooper, *An Introduction to Rehabilitation Engineering*. Boca Raton, FL: Taylor & Francis, 2007.
- [5] International Organization for Standardization, ISO 7176 series, *Wheelchairs* — in particular Part 1 (static stability), Part 2 (dynamic stability of electric wheelchairs), Part 5 (dimensions, mass and manoeuvring space), and Part 8 (static, impact and fatigue strengths). Geneva: ISO.
- [6] U.S. Department of Justice, *2010 ADA Standards for Accessible Design*, §307.4 (protruding objects) and §309.4 (operable parts). Washington, DC, 2010.
- [7] International Electrotechnical Commission, *IEC Guide 71: Guidelines for Standards Developers to Address the Needs of Older Persons and Persons with Disabilities*. Geneva: IEC, 2001.
- [8] International Organization for Standardization, *ISO 14971: Medical Devices — Application of Risk Management to Medical Devices*. Geneva: ISO.
- [9] U.S. Food and Drug Administration, *Applying Human Factors and Usability Engineering to Medical Devices: Guidance for Industry and Food and Drug Administration Staff*. February 3, 2016.
- [10] L. Demers, R. Weiss-Lambrou, and B. Ska, "The Quebec User Evaluation of Satisfaction with Assistive Technology (QUEST 2.0): An overview and recent progress," *Technology and Disability*, vol. 14, no. 3, pp. 101-105, 2002.